DATA 3464: Fundamentals of Data Processing

Intro to the course

Charlotte Curtis

September 3, 2026

Meet your instructor

Name: Charlotte Curtis

Pronouns: She/her

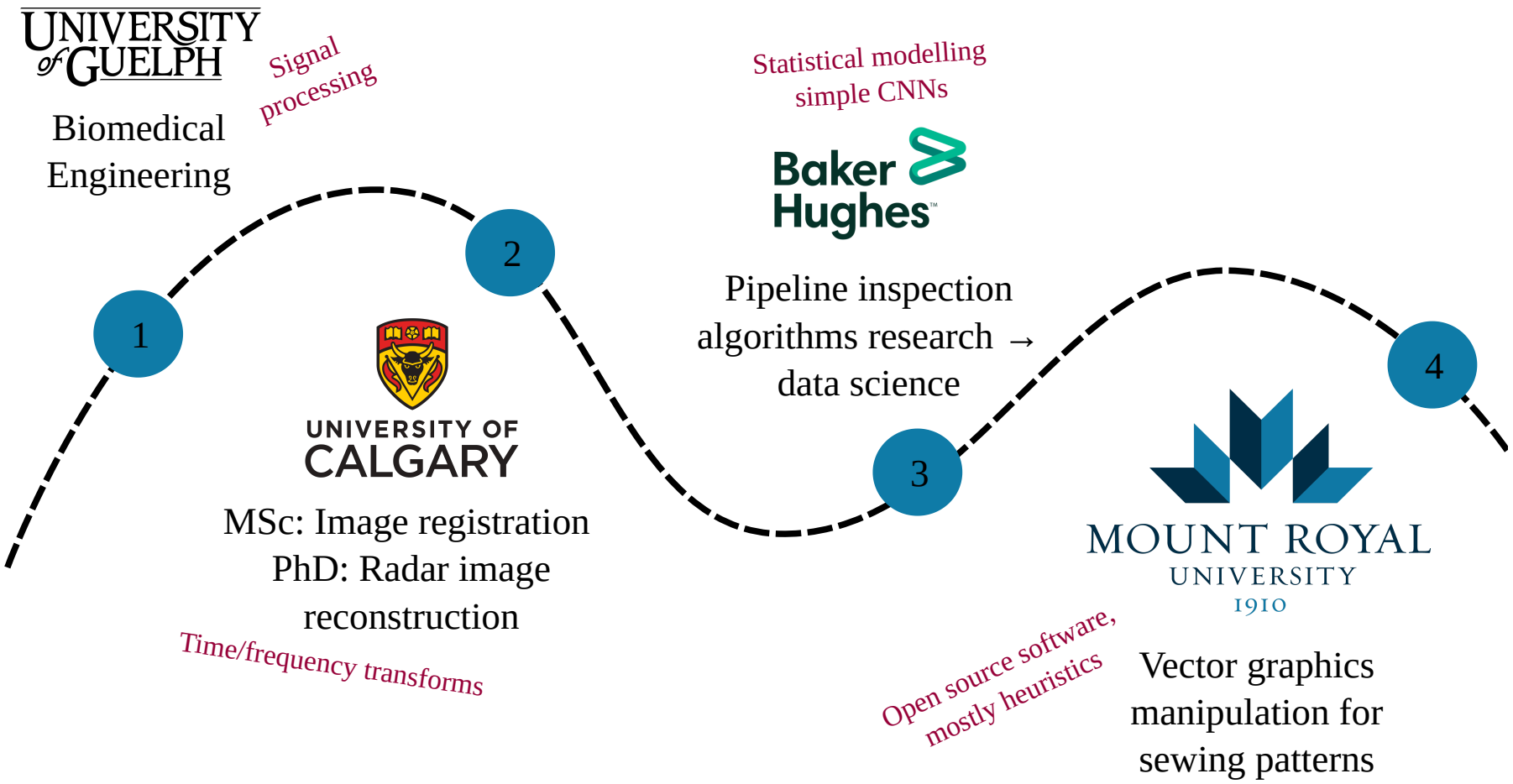
Office: B102-4

Email: ccurtis@mtroyal.ca

Office hours: Book here



My Background



Data processing

> *This course introduces techniques for ethically and responsibly wrangling and manipulating datasets to make them appropriate for addressing the question at hand. Topics may include cleaning and transforming data, integrity and quality measures, common file formats, feature selection and engineering, and generating features from unstructured sources such as text and images.*

Colloquially and lovingly referred to as **data wrangling**

Grade Assessment

Component	Weight	Comment
Tutorial exercises	10%	Pass/fail
Assignments	35%	8/12/15 split, in groups
Written tests	35%	x2, split 15%/20% in your favour
Final exam	20%	In lab with limited internet

> *A passing grade (50%) must be earned on the weighted average of written tests and final exam to be eligible for a C- in the course*

Course Website

Lecture notes, lab and assignment instructions will be hosted on an external site

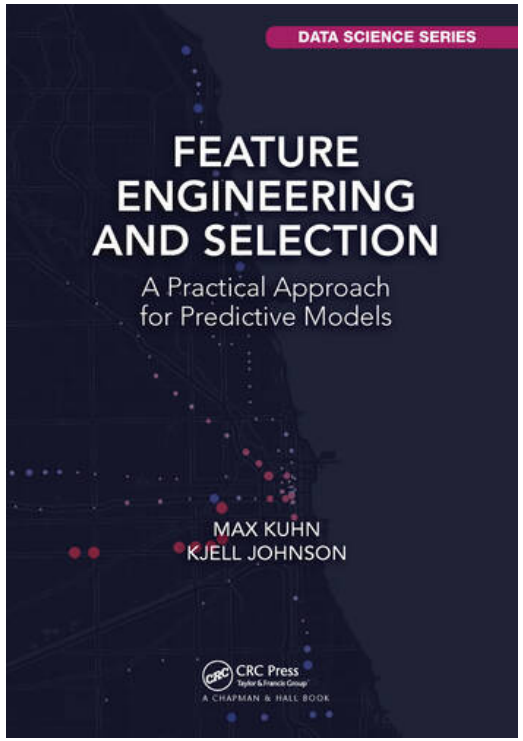
Rendered at: https://mru-data3464.github.io/f26

Source repo: https://github.com/mru-data3464/f26

> *Bonus marks may be awarded for substantial corrections to materials, submitted as pull requests*

Textbook(s)

- http://www.feat.engineering, additional open source resources as needed
- All the documentation!
  - Pandas
  - Numpy
  - SciPy
  - Scikit Learn
  - Matplotlib



> *Don't rely on AI summaries!*

Speaking of AI...

In this course (and others, and your career), you will need to know:

- **What** to do, and **why**
- **How** to do it

Note: AI will not be available during the final practical exam

> *Which of these things seem appropriate for AI assistance?*

Tentative weekly plan: Before reading week

Week	Topic	Chapter (ish)
0-1	Introduction and overview	1-2
1-2	Basic machine learning models and categorical encoding	3, 5
2-3	Exploring, splitting, and sampling data	4
4	Basic numeric data transformations	6
5	Dealing with missing and weird data	8

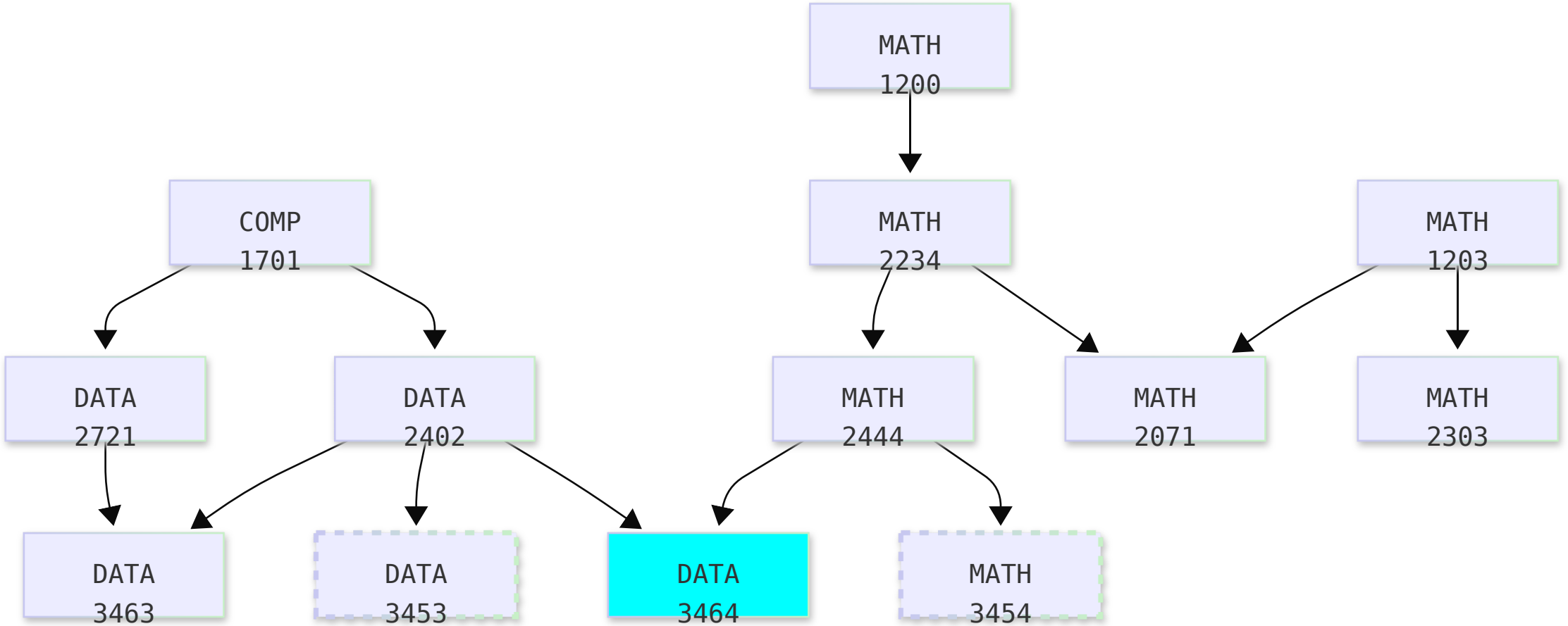
Written test 1: Thursday, Oct 7

Tentative weekly plan: second half

Week	Topic	Chapter (ish)
6	Fancy numeric and categorical transformations	5, 6
7	Wrangling text	
9	Bash scripting and data cards	
10	1D signals and audio	
11	Images and videos (2D signals)	
12	Presentations and buffer time	

Written test 2: Tuesday, Dec 1

Core courses so far



What do you know about...

- Various probability distributions
- Linear and logistic regression
- Data quality measures
- Data stewardship best practices
- Document parsing, web scraping, audio/video feature detection
- Linear algebra and array programming
- Prediction tasks: classification and regression
- Clustering and anomaly detection
- Evaluation metrics
- Basic data visualization (scatter plots, histograms, etc)

What do you want to know about?

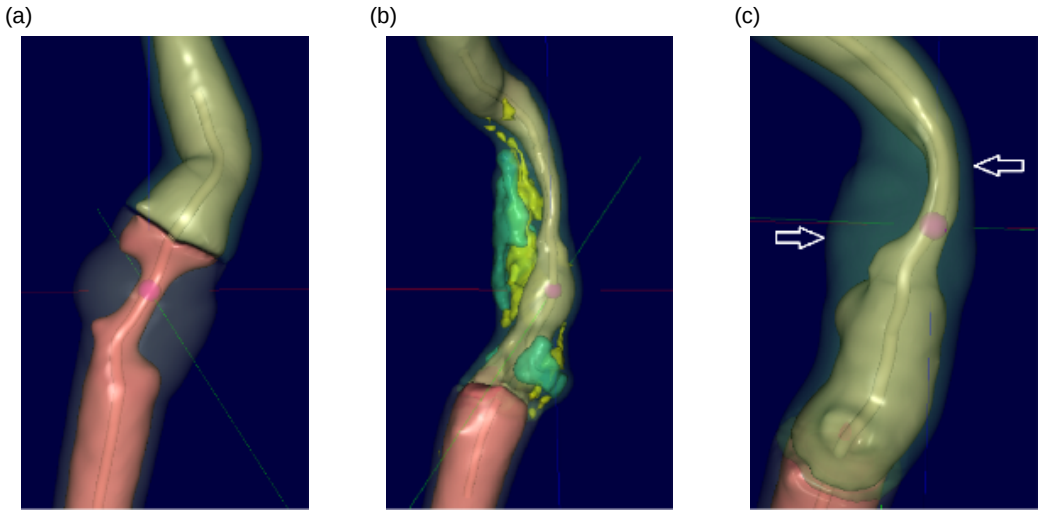
Examples of Subject Matter

- Finance
- Real estate
- Transportation
- Climate
- Politics
- Biology
- Chemistry
- Malware

Examples of Data types

- Unstructured text
- Structured text (e.g. csv, HTML)
- PDF
- Word documents
- Images
- Audio
- Video

Case study: risk of ischemic stroke



- Arterial stenosis can predict risk
- Plaque composition plays a role
- Features extracted from CT images
- Other risk factors (demographics, lifestyle) added to dataset

Chapter 2:

http://www.feat.engineering/stroke-tour

> *Many decisions in the data analysis process are subjective - I will often make different decisions than the textbook*

From data to prediction

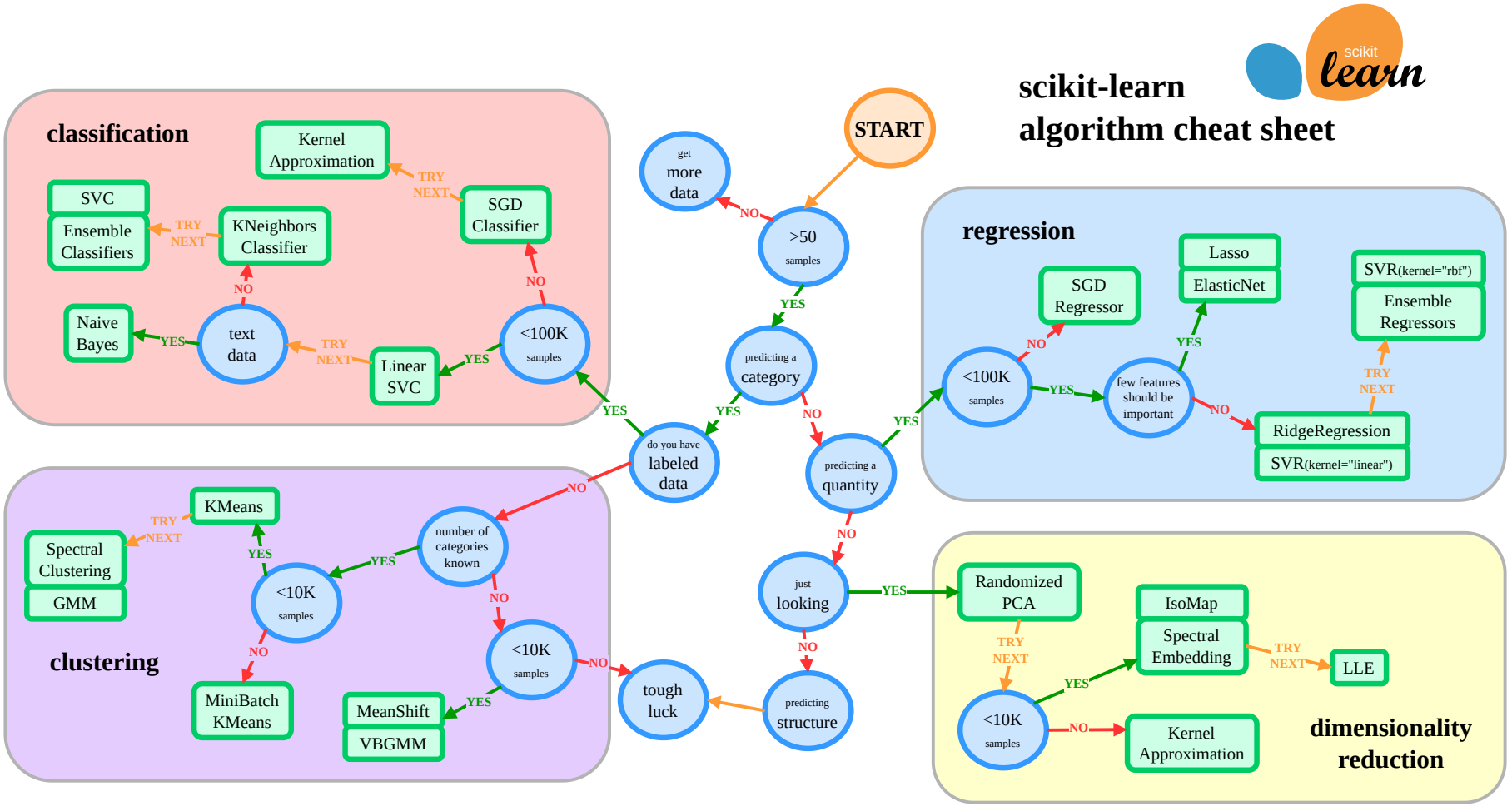
1. Understand the problem and define the task
2. Collect, anonymize and organize the data
3. Extract features
4. Explore the dataset
5. Select a model and preprocess
6. Train the model
7. Evaluate, fine-tune, iterate
8. Deploy and maintain your system

Adapted from Aurélien Géron's ML Project Checklist

Applied to the stroke example

1. What is the problem? What do we need to do?
2. (Collect, anonymize and organize the data) - Done for us
3. (Extract features) - Done for us
4. **Explore the dataset**
    - A critically important component, DO NOT OFFLOAD TO AI
    - This can even be where the data sciencing stops and we jump straight to visualizations and communicating insights!
    - Check out Data for Good case studies

5. Select a model and preprocess



Source: scikit-learn documentation

7. Evaluate, fine-tune, and iterate

- In my example, I jumped straight to testing on the held-back test set
- This is a terrible idea! We have no confidence that the model actually worked. We could be:
  - overfitting to the training data
  - making incorrect assumptions about the data
  - applying inappropriate transformations, or missing some
  - using the wrong model altogether

> *Validation needs to happen before the final testing*

Coming up next

- Lab: basic regression, show me where you're at
- Lecture: Simple models and evaluation metrics